Lecture 17

Unitary induction

Objectives. *Lecture 12 constructed induced representations of compact groups using an invariant measure on K/H. For a noncompact group G, the quotient G/H may carry no invariant measure. We replace it by a* quasi-invariant *measure, which translation by an element of G multiplies by a positive continuous function. We then correct the action by a square-root factor. The model is the $ax+b$ group, with representations induced from characters of its translation subgroup.*

By the end of this lecture you will be able to:

- State the quotient-measure theorem and derive the quasi-invariance factor and its cocycle identity.
- Define the normalised induced representation and prove that it is unitary.
- Compute the $ax+b$ representations induced from translation characters in two measure pictures.
- Explain why the unitary equivalence class of the induced representation depends only on the scaling orbit of the frequency.

17.1 Equivariant functions

Throughout, G is a Lie group, H is a closed subgroup and $q(x) = xH$ is the quotient map $G \to G/H$. By Lecture 3, G/H is a manifold on which G acts by $g \cdot xH = gxH$. The map q is continuous and open. Let (σ, V) be a unitary representation of H on a Hilbert space V.

Definition 17.1. An *equivariant function of type σ* is a continuous map $f : G \to V$ with

$$f(xh) = \sigma(h)^{-1} f(x) \qquad (x \in G,\ h \in H). \tag{17.1}$$

Let $\mathcal{F}_c(\sigma)$ be the space of those f for which $q(\operatorname{supp} f)$ is compact.

This is the convention of Lecture 12. Let $s : U \to G$ be a continuous section of q over an open set $U \subseteq G/H$. Then $f(s(p)h) = \sigma(h)^{-1} f(s(p))$. So over U the function f is just the ordinary function $f \circ s : U \to V$. Continuous local sections exist near every point; they are part of the construction of G/H in Lecture 3. Geometrically f is a section of a vector bundle over G/H. We use only the function description.

Since $\sigma(h)$ is unitary, $n_f(xH) = \|f(x)\|^2$ is well defined. It is continuous on G/H because q is open, and its support is compact.

We check that $\mathcal{F}_c(\sigma)$ is not zero when $V \neq 0$. Averaging over H turns compactly supported continuous functions $G \to V$ into elements of $\mathcal{F}_c(\sigma)$. Fix a left Haar measure dh on H. For continuous compactly supported $F\colon G \to V$ define $P_\sigma F$ by

$$P_\sigma F(x) = \int_H \sigma(h) F(xh)\, dh.$$

The integral is defined weakly by the Riesz argument of Lecture 10, applied to a continuous Hilbert-space-valued function with compact support. For fixed x, the integrand vanishes outside $H \cap x^{-1} \operatorname{supp} F$.

Left invariance of dh under $h' = h_0 h$ gives

$$P_\sigma F(xh_0) = \int_H \sigma(h_0^{-1}h') F(xh')\, dh' = \sigma(h_0)^{-1} P_\sigma F(x).$$

For x in a compact neighbourhood, all the integrands have support in one compact subset of H. Uniform continuity there makes $P_\sigma F$ continuous. Also $q(\operatorname{supp} P_\sigma F) \subseteq q(\operatorname{supp} F)$, which is compact. Thus $P_\sigma F \in \mathcal{F}_c(\sigma)$.

This element can be nonzero. Let $v \in V$ be nonzero. By strong continuity of σ, we can choose $\beta \in C_c(G)$ with $\beta \geq 0$ and $\beta(e) > 0$, supported so near e that $\operatorname{Re}\langle v, \sigma(h)v\rangle > 0$ for $h \in H \cap \operatorname{supp}\beta$. For $F(x) = \beta(x)v$ the number $\langle v, P_\sigma F(e)\rangle = \int_H \beta(h)\langle v, \sigma(h)v\rangle\, dh$ has positive real part. So $\mathcal{F}_c(\sigma) \neq 0$.

17.2 Quotient measures

Rescale a function on the line: $u \mapsto u((t - b_0)/a_0)$ with $a_0 > 0$. This multiplies $\|u\|^2$ by a_0, because dt is not invariant under dilation. Dividing by $a_0^{1/2}$ repairs the norm. This lecture carries out that repair for any quotient G/H.

In this lecture a *measure* on a locally compact space X means a positive Radon measure with full support, in the terms of Lecture 16. Equivalently, it is a positive linear functional $\varphi \mapsto \int_X \varphi\, d\mu$ on $C_c(X)$ which is strictly positive on nonzero $\varphi \geq 0$. Haar measures have this property. Then $L^2(X, \mu)$ is the completion of $C_c(X)$ for $\langle \varphi, \psi\rangle = \int_X \bar{\varphi}\psi\, d\mu$.

Fix left Haar measures dx on G and dh on H. Modular functions follow Lecture 16:

$$\int_G F(xg)\, dx = \Delta_G(g)^{-1} \int_G F(x)\, dx,$$

and likewise for Δ_H. For $F \in C_c(G)$ define $F^H \in C_c(G/H)$ by

$$F^H(xH) = \int_H F(xh)\, dh. \tag{17.2}$$

It is well defined by left invariance of dh.

Lemma 17.2. *If a measure μ on G/H satisfies $\int_G F\, dx = \int_{G/H} F^H\, d\mu$ for all $F \in C_c(G)$, then $\Delta_G(h) = \Delta_H(h)$ for all $h \in H$.*

Proof. Fix $h_0 \in H$ and apply the identity to $F_{h_0}(x) = F(xh_0)$. The left side becomes $\Delta_G(h_0)^{-1}\int_G F\,dx$. On the right,

$$(F_{h_0})^H(xH) = \int_H F(xhh_0)\,dh = \Delta_H(h_0)^{-1}F^H(xH).$$

Hence $\Delta_G(h_0)^{-1}\int_G F\,dx = \Delta_H(h_0)^{-1}\int_G F\,dx$. Choose $F \geq 0$ nonzero, so that $\int_G F\,dx > 0$. Dividing both sides by $\int_G F\,dx$ gives $\Delta_G(h_0)^{-1} = \Delta_H(h_0)^{-1}$, so $\Delta_G(h_0) = \Delta_H(h_0)$. ∎

For the $ax+b$ group and its scaling subgroup A, we will see $\Delta_G(a,0) = 1/a$ while $\Delta_A = 1$. Hence, by the modular compatibility lemma, Lemma 17.2, G/A carries no such measure. Instead, we insert a weight function on G into the integral over G.

The weight must transform in a definite way. Suppose that a continuous function $\rho > 0$ and a measure μ satisfy $\int_G F\rho\,dx = \int_{G/H} F^H\,d\mu$ for all $F \in C_c(G)$. Apply this identity to $F_{h_0}(x) = F(xh_0)$. As in the proof of Lemma 17.2, the right side is multiplied by $\Delta_H(h_0)^{-1}$. On the left, the modular identity applied to $x \mapsto F(x)\rho(xh_0^{-1})$ gives

$$\int_G F(xh_0)\rho(x)\,dx = \Delta_G(h_0)^{-1}\int_G F(x)\rho(xh_0^{-1})\,dx.$$

Since F is arbitrary, $\Delta_G(h_0)^{-1}\rho(xh_0^{-1}) = \Delta_H(h_0)^{-1}\rho(x)$ for all $x \in G$. Replacing x by xh_0 gives the transformation law in part (a) of the next theorem.

Theorem 17.3 (Quotient measures; cited). *Let G be a Lie group and H a closed subgroup.*

(a) There is a continuous function $\rho\colon G \to (0,\infty)$ with

$$\rho(xh) = \frac{\Delta_H(h)}{\Delta_G(h)}\,\rho(x) \qquad (x \in G,\ h \in H). \tag{17.3}$$

(b) The map $F \mapsto F^H$ sends $C_c(G)$ onto $C_c(G/H)$. If $\varphi \geq 0$, a lift F with $F^H = \varphi$ can be chosen with $F \geq 0$.

(c) For each ρ as in (a) there is a unique measure μ_ρ on G/H with

$$\int_G F(x)\rho(x)\,dx = \int_{G/H} F^H\,d\mu_\rho \qquad (F \in C_c(G)). \tag{17.4}$$

For a general closed subgroup, see G. B. Folland, *A Course in Abstract Harmonic Analysis*, second edition (2016), Chapter 2, “Measures on Homogeneous Spaces”, cited below as Folland, Abstract Harmonic Analysis. Its modular convention agrees with ours. Uniqueness of μ_ρ follows from the onto property of the averaging map. A. A. Kirillov, *Elements of the Theory of Representations* (1976), Section 9.1, cited below as Kirillov, Elements, treats quasi-invariant measures and the modular criterion. Its conventions must be translated before use.

A continuous positive function ρ with the transformation law (17.3) is called a *rho-function*. If $\Delta_G|_H = \Delta_H$ we may take $\rho = 1$. In this lecture ρ denotes a rho-function, and λ below denotes its cocycle. They are not the regular representations of Lecture 16.

Proposition 17.4. *Let ρ be a rho-function and $\mu = \mu_\rho$. Define λ by $\lambda(g, xH) = \rho(gx)/\rho(x)$. Then λ is well defined. For $g, g_1, g_2 \in G$, $p \in G/H$ and $\varphi \in C_c(G/H)$,*

$$\lambda(g_1 g_2, p) = \lambda(g_1, g_2 p)\,\lambda(g_2, p), \qquad \lambda(e, p) = 1, \tag{17.5}$$

$$\int_{G/H} \varphi(g^{-1}p)\,d\mu(p) = \int_{G/H} \varphi(p)\,\lambda(g, p)\,d\mu(p). \tag{17.6}$$

Proof. By the transformation law (17.3), replacing x by xh multiplies $\rho(gx)$ and $\rho(x)$ by the same factor. So λ is well defined. The function $(g, x) \mapsto \rho(gx)/\rho(x)$ is continuous and constant on fibres of $\mathrm{id} \times q$. Since q is open, $\mathrm{id} \times q$ is an open surjection, hence a quotient map. Thus λ is jointly continuous in (g, p). For $p = xH$,

$$\lambda(g_1 g_2, p) = \frac{\rho(g_1 g_2 x)}{\rho(g_2 x)} \cdot \frac{\rho(g_2 x)}{\rho(x)} = \lambda(g_1, g_2 p)\,\lambda(g_2, p).$$

Also $\lambda(e, p) = \rho(x)/\rho(x) = 1$.

For the quasi-invariance identity (17.6), use the onto property of averaging, Theorem 17.3(b), to write $\varphi = F^H$ with $F \in C_c(G)$. The function $x \mapsto F(g^{-1}x)$ has average $\varphi(g^{-1}\,\cdot)$. By the weighted quotient integral (17.4) and left invariance of dx, with $y = g^{-1}x$,

$$\int \varphi(g^{-1}p)\,d\mu(p) = \int_G F(g^{-1}x)\rho(x)\,dx = \int_G F(y)\rho(gy)\,dy = \int_G F(y)\,\lambda(g, yH)\,\rho(y)\,dy.$$

Since $\lambda(g, \cdot)$ is constant on cosets, $y \mapsto F(y)\lambda(g, yH)$ has average $\varphi\,\lambda(g, \cdot)$. Applying the weighted quotient integral (17.4) again gives the right side of (17.6). ∎

We say μ is *quasi-invariant*: translation by g changes it by the positive continuous factor $\lambda(g, \cdot)$. If $\rho = 1$, then $\lambda = 1$ and μ is G-invariant.

17.3 The normalised induced representation

The substitution $p = gq$ in the quasi-invariance identity (17.6) gives $d\mu(gq) = \lambda(g, q)\,d\mu(q)$. So $\lambda(g, \cdot)$ is the Jacobian of $q \mapsto gq$ relative to μ, and the cocycle identity (17.5) is the chain rule. As an analogy, think of $\|f\|^2\,d\mu$ as a probability density. Then an amplitude carries the square root of the Jacobian.

Fix a rho-function ρ, and let μ and λ be as in the quasi-invariance proposition, Proposition 17.4. For $f_1, f_2 \in \mathcal{F}_c(\sigma)$ put

$$\langle f_1, f_2 \rangle = \int_{G/H} \langle f_1(x), f_2(x) \rangle\,d\mu(xH), \qquad \|f\|^2 = \int_{G/H} n_f\,d\mu.$$

The integrand is well defined because σ is unitary. If $\|f\| = 0$, then $n_f = 0$ by strict positivity, so $f = 0$.

Theorem 17.5. *For $f \in \mathcal{F}_c(\sigma)$ and $g \in G$ define*

$$(\Pi(g)f)(x) = \left(\frac{\rho(g^{-1}x)}{\rho(x)}\right)^{1/2} f(g^{-1}x) = \lambda(g^{-1}, xH)^{1/2} f(g^{-1}x). \tag{17.7}$$

Then (a) $\Pi(g)f \in \mathcal{F}_c(\sigma)$; (b) $\Pi(e) = I$ *and* $\Pi(g_1 g_2) = \Pi(g_1)\Pi(g_2)$; (c) $\|\Pi(g)f\| = \|f\|$.

Proof. (a) The factor depends only on xH. So replacing x by xh only changes $f(g^{-1}x)$ into $\sigma(h)^{-1}f(g^{-1}x)$. Also $q(\operatorname{supp}\Pi(g)f) = g \cdot q(\operatorname{supp} f)$ is compact.

(b) Let $p = xH$. Applying the normalised action (17.7) twice,

$$(\Pi(g_1)\Pi(g_2)f)(x) = \lambda(g_1^{-1}, p)^{1/2}\,\lambda(g_2^{-1}, g_1^{-1}p)^{1/2}\,f(g_2^{-1}g_1^{-1}x).$$

The cocycle identity for the pair g_2^{-1}, g_1^{-1} turns the product of factors into $\lambda((g_1g_2)^{-1}, p)^{1/2}$. Also $\lambda(e, p) = 1$ gives $\Pi(e) = I$.

(c) Translation by g changes μ by the factor $\lambda(g, \cdot)$. The square-root factor in $\Pi(g)$ is chosen to cancel this change. We have $n_{\Pi(g)f}(p) = \lambda(g^{-1}, p)\,n_f(g^{-1}p)$. Put $\varphi(p) = n_f(p)\,\lambda(g^{-1}, gp)$, so that $\varphi(g^{-1}p) = n_{\Pi(g)f}(p)$. Then the quasi-invariance identity (17.6) gives

$$\|\Pi(g)f\|^2 = \int \varphi(g^{-1}p)\,d\mu = \int n_f(p)\,\lambda(g^{-1}, gp)\,\lambda(g, p)\,d\mu(p) = \int n_f\,d\mu.$$

The last step is the cocycle identity (17.5) with $g_1 = g^{-1}$ and $g_2 = g$. ∎

By the representation law and the isometry property in the normalised-induction theorem, Theorem 17.5, each $\Pi(g)$ is an isometry of $\mathcal{F}_c(\sigma)$ with inverse $\Pi(g^{-1})$. So $\Pi(g)$ extends to a unitary operator on the completion. The representation law persists by continuity.

Definition 17.6. The *normalised induced representation* $\operatorname{Ind}_H^G \sigma$ is the completion of $\mathcal{F}_c(\sigma)$, for the inner product defined by the chosen rho-function ρ, together with these unitary operators $\Pi(g)$.

Proposition 17.7. *The induced representation* $\operatorname{Ind}_H^G \sigma$ *is strongly continuous.*

Proof. First take $f \in \mathcal{F}_c(\sigma)$ and let $C = q(\operatorname{supp} f)$, a compact set. Choose a compact neighbourhood B of e in G. For $g \in B$, the quotient supports of both $\Pi(g)f$ and f lie in the compact set

$$S = B \cdot C \subset G/H.$$

It suffices to show that their difference tends uniformly to zero in fibre norm on S. This norm is independent of the representative in G because σ is unitary.

By compactness of S and the manifold structure of G/H, we can cover S by finitely many compact subsets S_i. Each S_i lies in an open set with a continuous local section s_i of q. The lifted sets $s_i(S_i)$ are compact. At these representatives,

$$(\Pi(g)f - f)(s_i(p)) = \lambda(g^{-1}, p)^{1/2} f(g^{-1}s_i(p)) - f(s_i(p)).$$

The function $(g, p) \mapsto \lambda(g^{-1}, p)$ is jointly continuous and equals one at $g = e$. Also $(g, p) \mapsto f(g^{-1}s_i(p))$ is jointly continuous as a Hilbert-space-valued function. On the compact sets $B \times S_i$, this continuity is uniform. Thus the displayed difference tends uniformly to zero for $p \in S_i$ as $g \to e$. There are only finitely many S_i, so the same holds on S.

Choose $\varphi \in C_c(G/H)$ with $\varphi \geq 0$ and $\varphi \geq 1$ on S. If the uniform bound is ε, then

$$\|\Pi(g)f - f\|^2 \leq \varepsilon^2 \int_{G/H} \varphi\,d\mu.$$

This proves that $\Pi(g)f \to f$ in the norm of the completion as $g \to e$, for each $f \in \mathcal{F}_c(\sigma)$.

For a vector v in the completion, choose f with $\|v - f\| < \varepsilon$. Unitarity gives

$$\|\Pi(g)v - v\| \le 2\varepsilon + \|\Pi(g)f - f\|.$$

Hence continuity holds at e for every v. By the representation law and unitarity, continuity at e implies continuity at every g_0. ∎

Remark 17.8. Without the square root, the map $f \mapsto f(g^{-1}\cdot)$ still preserves $\mathcal{F}_c(\sigma)$ and satisfies the representation law. By the quasi-invariance identity (17.6), this map sends $\|f\|^2$ to $\int n_f\,\lambda(g,\cdot)\,d\mu$. So in general it is not unitary; in the $ax+b$ example it multiplies $\|\psi\|^2$ by a_0 (Proposition 17.9(a)). For compact G and finite-dimensional σ, we take $\rho = 1$ and recover the construction of Lecture 12. Different rho-functions give unitarily equivalent representations (Exercise 17.3). Finite-group Frobenius reciprocity does not transfer verbatim. For example, the representation Π_0 of § 17.5 is induced from the trivial character of the translation subgroup N. A character of the $ax+b$ group that is trivial on N would have a nonzero N-map to that trivial character, yet no such character occurs in Π_0. We do not use reciprocity.

17.4 Representations of the $ax + b$ group induced from translations

Let $G = \{(a, b) : a > 0,\ b \in \mathbb{R}\}$ with

$$(a,b)(a',b') = (aa', b + ab'), \qquad (a,b)^{-1} = (a^{-1}, -a^{-1}b), \qquad (a,b) = \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}.$$

It acts on $\mathbb{R}$ by $t \mapsto at + b$. By Lecture 16, $dx = a^{-2}\,da\,db$ is a left Haar measure and $\Delta_G(a,b) = 1/a$. Let $N = \{n_c = (1,c)\}$ be the translations and $A = \{(a,0)\}$ the scalings. Then $N \cong \mathbb{R}$ has Haar measure dc and $\Delta_N = 1$. For a *frequency* $\xi \in \mathbb{R}$, the map $\chi_\xi(n_c) = e^{i\xi c}$ is a unitary character of N.

The quotient G/N carries an invariant measure, as we see below. We treat it first because its points carry the orbit of the frequency. The affine line G/A of the opening preview has no invariant measure, and it comes at the end of § 17.5.

17.4.1 The quotient and two measures

Since $(a,b)n_c = (a, b + ac)$, the coset $(a,b)N$ is $\{a\} \times \mathbb{R}$. So $(a,b)N \mapsto a$ identifies G/N with $(0,\infty)$, and $s(a) = (a,0)$ is a global section. The action is $(a_0,b_0)\cdot a = a_0 a$.

Since $\Delta_G(n_c) = 1 = \Delta_N(n_c)$, the transformation law (17.3) says $\rho(xn) = \rho(x)$. Thus rho-functions are positive continuous functions of a alone. The substitution $b = ac$ gives

$$F^N(a) = \int_{\mathbb{R}} F(a,ac)\,dc = \frac{1}{a}\int_{\mathbb{R}} F(a,b)\,db.$$

Hence for $\rho = \rho(a)$,

$$\int_G F\rho\,dx = \int_0^\infty \Big(\int_{\mathbb{R}} F(a,b)\,db\Big)\frac{\rho(a)\,da}{a^2} = \int_0^\infty F^N(a)\,\rho(a)\,\frac{da}{a}.$$

By the uniqueness in the quotient-measure theorem, Theorem 17.3, $d\mu_\rho = \rho(a)\,da/a$.

We use two choices. The rho-function $\rho_1 = 1$ gives the invariant measure, and the rho-function $\rho_{\mathrm{Leb}}(a,b) = a$ gives Lebesgue measure:

$$d\mu_{\rho_1} = \frac{da}{a}, \qquad d\mu_{\rho_{\mathrm{Leb}}} = da.$$

For the Lebesgue choice, take $g = (a_0, b_0)$ and $x = (a, 0)$. Then $gx = (a_0 a, b_0)$, so $\lambda_{\mathrm{Leb}}(g, a) = a_0$. In this case the quasi-invariance identity (17.6) is the substitution rule $\int_0^\infty \varphi(a/a_0)\, da = a_0 \int_0^\infty \varphi(u)\, du$. The cocycle identity reads $a_1 a_2 = a_1 \cdot a_2$, because $g_1 g_2$ has first entry $a_1 a_2$.

17.4.2 Equivariant functions and the action

For $\sigma = \chi_\xi$, the equivariance condition (17.1) says $f(a, b + ac) = e^{-i\xi c} f(a, b)$. Take $b = 0$ and then $c = b/a$:

$$f(a,b) = e^{-i\xi b/a}\varphi(a), \qquad \varphi = f \circ s. \tag{17.8}$$

Conversely, for $\varphi \in C_c((0,\infty))$ this f lies in $\mathcal{F}_c(\chi_\xi)$, since $e^{-i\xi(b+ac)/a} = e^{-i\xi c} e^{-i\xi b/a}$. So $f \mapsto \varphi$ is a bijection onto $C_c((0,\infty))$. Moreover $n_f = |\varphi|^2$ and $\|f\|^2 = \int_0^\infty |\varphi|^2 \rho(a)\, da/a$.

Figure 17.1 draws such a function. Left translation by n_c sends (a, b) to $(a, b + c)$. So on the coset over a, the function f oscillates in b with frequency ξ/a. In the coset parameter $c = b/a$ the frequency is ξ.

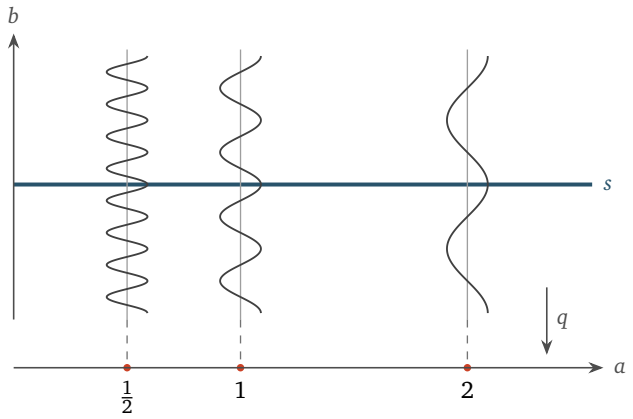


Figure 17.1: The real part of an equivariant function $f(a,b) = e^{-i\xi b/a}\varphi(a)$ with $\xi = 2\pi$, drawn as a horizontal offset on the three cosets of N over $a = \frac{1}{2}, 1, 2$. The heavy line is the section s, and q sends each vertical coset to its point of the base $(0,\infty)$.

Let $g = (a_0, b_0)$. Then $g^{-1}(a, 0) = (a/a_0,\ -b_0/a_0)$, and the equivariant form (17.8) gives

$$f(g^{-1}(a,0)) = \exp\Big(-i\xi\,\frac{-b_0/a_0}{a/a_0}\Big)\varphi\Big(\frac{a}{a_0}\Big) = e^{i\xi b_0/a}\,\varphi\Big(\frac{a}{a_0}\Big).$$

The factor at $x = (a, 0)$ is $(\rho(a/a_0)/\rho(a))^{1/2}$. This yields two pictures:

$$(\Pi_\xi(a_0,b_0)\varphi)(a) = e^{i\xi b_0/a}\,\varphi(a/a_0) \qquad \text{on } L^2((0,\infty), da/a), \tag{17.9}$$

$$(\Pi^{\mathrm{Leb}}_\xi(a_0,b_0)\psi)(a) = a_0^{-1/2} e^{i\xi b_0/a}\,\psi(a/a_0) \qquad \text{on } L^2((0,\infty), da). \tag{17.10}$$

The normalised-induction theorem, Theorem 17.5, already gives unitarity. We now verify unitarity and the representation law directly in this model.

Proposition 17.9. *Let $g = (a_0, b_0)$, $g_j = (a_j, b_j)$ and $\varphi, \psi \in C_c((0,\infty))$.*

(a) $\|\Pi^{\mathrm{Leb}}_\xi(g)\psi\| = \|\psi\|$, *while* $\psi \mapsto e^{i\xi b_0/a}\psi(a/a_0)$ *multiplies* $\|\psi\|^2$ *by* a_0.

(b) $\Pi^{\mathrm{Leb}}_\xi(g_1)\Pi^{\mathrm{Leb}}_\xi(g_2) = \Pi^{\mathrm{Leb}}_\xi(g_1 g_2)$.

(c) $(W\varphi)(a) = a^{-1/2}\varphi(a)$ *extends to a unitary map* $L^2(da/a) \to L^2(da)$ *with* $W\Pi_\xi(g) = \Pi^{\mathrm{Leb}}_\xi(g)W$.

Proof. (a) Substitute $a = a_0 u$:

$$\int_0^\infty a_0^{-1}|\psi(a/a_0)|^2\,da = \int_0^\infty |\psi(u)|^2\,du.$$

Without the factor a_0^{-1} the result is $a_0\|\psi\|^2$.

(b) Since $g_1 g_2 = (a_1 a_2,\ b_1 + a_1 b_2)$,

$$\begin{aligned}(\Pi^{\mathrm{Leb}}_\xi(g_1)\Pi^{\mathrm{Leb}}_\xi(g_2)\psi)(a) &= a_1^{-1/2}e^{i\xi b_1/a}\,(\Pi^{\mathrm{Leb}}_\xi(g_2)\psi)(a/a_1)\\ &= a_1^{-1/2}a_2^{-1/2}\,e^{i\xi b_1/a}e^{i\xi a_1 b_2/a}\,\psi\Big(\frac{a}{a_1 a_2}\Big)\\ &= (a_1 a_2)^{-1/2}e^{i\xi(b_1+a_1 b_2)/a}\,\psi\Big(\frac{a}{a_1 a_2}\Big) = (\Pi^{\mathrm{Leb}}_\xi(g_1 g_2)\psi)(a).\end{aligned}$$

The square roots combine because λ_{Leb} is multiplicative; this is the cocycle identity. The phases combine into the second entry of $g_1 g_2$; this is where equivariance enters.

(c) We have $\int |a^{-1/2}\varphi|^2\,da = \int |\varphi|^2\,da/a$, and W has inverse multiplication by $a^{1/2}$. Finally,

$$(\Pi^{\mathrm{Leb}}_\xi(g)W\varphi)(a) = a_0^{-1/2}e^{i\xi b_0/a}\Big(\frac{a}{a_0}\Big)^{-1/2}\varphi\Big(\frac{a}{a_0}\Big) = a^{-1/2}e^{i\xi b_0/a}\varphi\Big(\frac{a}{a_0}\Big) = (W\Pi_\xi(g)\varphi)(a). \qquad \blacksquare$$

Note that W is multiplication by $(\rho_1/\rho_{\mathrm{Leb}})^{1/2}$.

17.5 The scaling orbit of a frequency

On translations, the first picture (17.9) gives $(\Pi_\xi(n_c)\varphi)(a) = e^{i(\xi/a)c}\varphi(a)$. So at the point $a \in G/N$ the group N acts by the character $\chi_{\xi/a}$. Let A act on characters of N by $(t\cdot\chi)(n) = \chi(t^{-1}nt)$. For $t = (a, 0)$,

$$t^{-1}n_c t = (a^{-1}, a^{-1}c)(a, 0) = (1, c/a), \qquad \text{so} \qquad t\cdot\chi_\xi = \chi_{\xi/a}.$$

Thus the character at aN is $s(a)\cdot\chi_\xi$. As a runs through $(0,\infty)$, the frequency ξ/a fills the orbit $\mathbb{R}_{>0}\,\xi$. There are three orbits: $(0,\infty)$, $(-\infty, 0)$ and $\{0\}$.

Proposition 17.10. *Let $\kappa > 0$. The operator $(U\varphi)(a) = \varphi(a/\kappa)$ is unitary on $L^2((0,\infty), da/a)$, and $U\Pi_\xi(g) = \Pi_{\kappa\xi}(g)U$ for all $g \in G$.*

Proof. The substitution $a = \kappa u$ preserves da/a, so U is isometric on $C_c((0,\infty))$. Its inverse is $\varphi \mapsto \varphi(\kappa\,\cdot)$. For $g = (a_0, b_0)$,

$$(U\Pi_\xi(g)\varphi)(a) = (\Pi_\xi(g)\varphi)(a/\kappa) = e^{i\xi b_0\kappa/a}\,\varphi\Big(\frac{a}{\kappa a_0}\Big),$$
$$(\Pi_{\kappa\xi}(g)U\varphi)(a) = e^{i\kappa\xi b_0/a}\,(U\varphi)(a/a_0) = e^{i\kappa\xi b_0/a}\,\varphi\Big(\frac{a}{a_0\kappa}\Big).$$

The two expressions agree on the dense subspace $C_c((0,\infty))$ of $L^2((0,\infty), da/a)$. Both sides are bounded operators, so they agree on the whole space. ∎

Thus, by Proposition 17.10, the unitary equivalence class of Π_ξ depends only on the scaling orbit of ξ. We write $\Pi_+ = \Pi_1$ and $\Pi_- = \Pi_{-1}$. By Exercise 17.5, the three representations Π_+, Π_- and Π_0 are pairwise inequivalent.

In Π_0 we have $(\Pi_0(a_0, b_0)\varphi)(a) = \varphi(a/a_0)$. So N acts trivially. The representation Π_0 is the left regular representation of $(0,\infty) \cong G/N$, composed with the quotient map $G \to G/N$.

We do not prove irreducibility here. Mackey's theorem, cited in Lecture 25, applies with $n = 1$ and $H = (0,\infty)$. It shows that Π_+ and Π_- are irreducible. It also shows that, together with the characters $(a, b) \mapsto a^{is}$ with $s \in \mathbb{R}$, they are all the irreducible unitary representations of G, up to unitary equivalence. The representation Π_0 is not irreducible. By Exercise 17.4(a) it is the translation representation on $L^2(\mathbb{R})$, composed with $(a_0, b_0) \mapsto \log a_0$. By the proposition on finite-dimensional invariant subspaces in Lecture 15, it has no invariant line. An irreducible unitary representation of the abelian group $(0,\infty)$ is one-dimensional, by the Schur lemma that Lecture 18 cites. So Π_0 is reducible.

In physics this is the pattern of Wigner's classification. Translations act by phases, the remaining group moves the frequency along an orbit, and the representation lives on that orbit. Lecture 25 carries this out for the Poincaré group. Nothing in this lecture depends on it.

A quotient with no invariant Radon measure

For comparison, give the scaling subgroup $A = \{(a, 0) : a > 0\}$ the Haar measure da/a and induce its trivial character. Identify G/A with the line by $(a, b)A \mapsto b$. Left multiplication gives the familiar affine action

$$(a_0, b_0) \cdot t = b_0 + a_0 t.$$

Lebesgue measure dt is quasi-invariant, but a dilation multiplies it by a_0. Any invariant Radon measure would be translation invariant. By Haar uniqueness on $\mathbb{R}$, it would be a constant multiple of dt. Dilation invariance then forces this constant to be zero.

Here $\Delta_A = 1$ and $\Delta_G(a, 0) = 1/a$, so a rho-function is $\rho(a, b) = a$. The quotient identity follows directly:

$$\int_G F(a,b)\rho(a,b)\,\frac{da\,db}{a^2} = \int_{\mathbb{R}} \left(\int_0^\infty F(a,t)\,\frac{da}{a} \right) dt.$$

Since $(1, t)(a, 0) = (a, t)$ and A carries the Haar measure da/a, the inner integral is the coset average $F^A((1,t)A)$ of (17.2) with $H = A$. By the uniqueness in the quotient-measure theorem, Theorem 17.3, dt is exactly the measure μ_ρ supplied by this rho-function.

The square-root factor in the normalised action (17.7) becomes $a_0^{-1/2}$, and the induced action is

$$(\Pi(a_0, b_0)u)(t) = a_0^{-1/2} u\left(\frac{t - b_0}{a_0}\right).$$

To verify its norm, set $r = (t - b_0)/a_0$. Then $dt = a_0\, dr$, so

$$\|\Pi(a_0, b_0)u\|_2^2 = a_0^{-1} \int_{\mathbb{R}} |u((t - b_0)/a_0)|^2 dt = \int_{\mathbb{R}} |u(r)|^2 dr.$$

Without the square root, the squared norm would be multiplied by a_0. This quotient makes the measure correction visible in one variable. In the earlier translation-subgroup model, the quotient was the positive scaling line and an invariant quotient measure was available. The subgroup, not just the ambient group, controls the answer. Exercise 17.6 shows that this representation is unitarily equivalent to $\Pi_1 \oplus \Pi_{-1}$.

Self-checks

1. In the $ax + b$ group, derive the transformation law of a rho-function for $H = A$ and check that $\rho(a, b) = a$ satisfies it. Is $\rho = 1$ possible?
   Answer. For $h = (t, 0)$ we have $\Delta_A(h)/\Delta_G(h) = 1/(1/t) = t$, so we need $\rho(xh) = t\rho(x)$. Since $(a, b)(t, 0) = (at, b)$, the function $\rho(a, b) = a$ works. The choice $\rho = 1$ fails for $t \neq 1$.
2. Is $\rho(a, b) = a + b^2$ a rho-function for $H = N$?
   Answer. No. A rho-function for N must satisfy $\rho(xn) = \rho(x)$. But $\rho((a, 0)n_1) = \rho(a, a) = a + a^2 \neq a = \rho(a, 0)$.
3. Give $\Pi_\xi(g)^*$ for $g = (a_0, b_0)$ in the first picture (17.9).
   Answer. Since Π_ξ is unitary, the adjoint is $\Pi_\xi(g^{-1})$, with $g^{-1} = (a_0^{-1}, -a_0^{-1}b_0)$. Hence

   $$(\Pi_\xi(g)^*\varphi)(a) = e^{-i\xi b_0/(a_0 a)}\varphi(a_0 a).$$

   Substituting $a = a_0 u$ in $\langle \Pi_\xi(g)\varphi, \psi\rangle$ confirms this formula.

Summary

A rho-function gives a quasi-invariant measure μ_ρ on G/H. Its translation factor $\lambda(g, xH) = \rho(gx)/\rho(x)$ is a cocycle. The action $f \mapsto \lambda(g^{-1}, \cdot)^{1/2} f(g^{-1}\,\cdot)$ on equivariant functions is unitary, because the cocycle cancels the change of measure. For the representation Π_ξ of the $ax + b$ group induced from the character χ_ξ of the translation subgroup N, the point $a \in G/N$ carries the frequency ξ/a. The unitary equivalence class of Π_ξ depends only on the scaling orbit of ξ. By Mackey's theorem, cited in Lecture 25, Π_+ and Π_- are irreducible, while Π_0 is not.

Reading

Folland, Abstract Harmonic Analysis, Chapter 2, "Measures on Homogeneous Spaces": quotient measures and rho-functions for a general closed subgroup. Kirillov, Elements,

Section 9.1: quasi-invariant measures on homogeneous spaces. Kirillov, Elements, Section 13: models of induced representations. A. W. Knapp, *Representation Theory of Semisimple Groups* (2001), Sections VII.1–VII.2: pictures of representations induced from a parabolic subgroup of a linear connected reductive group.